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Fundamental structure forming phenomena of polycrystalline films and the structure zone models

P.B. Barna *, M. Adamik

Research Institute for Technical Physics of Hungarian Academy of Sciences, P.O. Box 76, Budapest H-1325, Hungary

Abstract

Previous structure zone models used for the interpretation of the experimental results were constructed by compiling the various structures found in thick polycrystalline films deposited at different temperatures and parameters. The present paper shows that while the temperature dependence of surface and bulk diffusion can not adequately describe the evolution mechanisms of various polycrystalline thin film structure, the fundamental structure forming phenomena, including the contribution of impurities introduced by the authors make possible the comprehensive description of the structure forming mechanisms. Real structure zone models including also the related textures, are constructed by considering the concentration of impurities (co-deposited additives) in the vapour beam as a new deposition parameter, and the fundamental structure forming phenomena. These are phenomena composed of atomic processes, and can give account for the global effect of the atomic processes and are directly responsible for the structure evolution. Impurities (additives) can promote or inhibit the operation of the structure forming phenomena (promoter or inhibitor resp.). The new structure zone models can be used to tailor polycrystalline film structures of large variety (from large grained through columnar to the micro- or nanocrystalline structures) and to diagnose technologies. © 1998 Elsevier Science S.A.

Keywords: Polycrystalline films; Temperature; Atomic

1. Introduction

Polycrystalline thin films prepared by various techniques at a large scale of parameters exhibit large variety of structures considering the size, morphology and orientation relationship of crystals as well as the microchemistry of the film. Structural inhomogeneity along the film thickness can also be present. The structure zone models (SZM) [1–7], constructed by the compilation of the experimental results, present an important guideline for systematising both the experimental results and the dependence of structures on the deposition temperature (considering the homologous temperature, T_s/T_m, where T_s is the substrate temperature and T_m is the melting point of the film material). It is generally agreed that the surface and bulk diffusion are the most determinant atomic processes of the structure evolution. These can, however, in themselves not explain the evolution of the various, sometimes peculiar structures and the effects of parameters on them completely.

The detailed analysis of the role of impurities in the modification of the conventional film structures and in the development of peculiar structural features [8–12] urged the authors to introduce the pressure or relative impingement rate of active impurities as an additional parameter of the film deposition [13,14]. It has been shown recently that by selecting so called fundamental structure forming phenomena a rather comprehensive and general description of the structure evolution (size and shape of grains together with their crystal orientation) in polycrystalline thin films can be given and an universal SZM can be constructed taking into consideration the effects of co-depositing impurity species [15]. (The co-deposition of some additives and dopants beyond their solubility limit, or component(s) of multicomponent films at concentrations higher than the required stoichiometry have similar effects on the structure evolution as those due to the active impurities, therefore, the expression 'impurity' will be used consistently in general meaning in this paper.)

The present paper is concentrating on some details of the structure evolution of metal thin films by considering the fundamental structure forming phenomena. It will be shown that the active impurities can not only inhibit but also promote the structure evolution in vapour deposited

* Corresponding author.

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polycrystalline thin films. The basic concepts of the universal structure zone model, describing both the morphology and texture, will also be discussed.

2. The fundamental structure forming phenomena

It is well known that the growth of thin films proceeds through consecutive stages characterised by specific processes of structure evolution: nucleation, island growth, coalescence of islands, formation of polycrystalline islands and channels, development of continuous structure and thickness growth. The evolution of the structure in polycrystalline thin films is a very complex phenomenon and exhibits different features in different stages of film growth.

The analysis of the structure evolution and that of the effects of the deposition parameters on the structure and on the appearance of peculiar structural features lead to the conclusion that the comprehensive description of the formation of the various structures is possible by selecting the *basic structure forming phenomena* well known in materials science and solid state physics (nucleation, crystal growth, grain growth), and an *additional phenomenon*, the process-induced segregation of foreign species related to crystal growth and grain growth [16–19]. These are the *fundamental structure forming phenomena* active during the structure evolution of real polycrystalline thin films [14]. These phenomena are composed of elementary atomic processes, and can give account for the global effects of the atomic processes on the structure evolution. It is important to note that the atomic processes are controlled directly not only by the temperature but also by the structural conditions characteristic of the actual growth stage (structural preconditions). It has a particular importance that, by considering the fundamental structure forming phenomena, both the evolution of the size and morphology of the grains as well as that of the orientation of crystals can be described.

The fundamental structure forming phenomena are summarised in the following.

2.1. The nucleation

Nucleation starting the growth of individual islands takes place on the substrate surface at the very first stage of the condensation (*primary nucleation*) or later on the bare substrate surface area developing upon liquid like coalescence (*secondary nucleation*). A peculiar case of nucleation shows up on the surface of a growing crystal when its growth is blocked by a surface covering layer (SCL) of an impurity phase. This is the *repeated nucleation* [16]. The primary nucleation starts the condensation and the film growth on the whole substrate surface simultaneously, while the secondary and the repeated nucleation initiates the starting of the growth locally in later stages of film formation. It is important to note that on amorphous substrates the nuclei are randomly oriented. The kinetics of nucleation are discussed in review articles in details [20–26].

2.2. The crystal growth

Crystal growth is the fundamental structure forming phenomenon which incorporates the depositing material into the condensed phase. Two main cases of crystal growth should be considered in case of polycrystalline thin films: the growth of discrete crystals dispersed on the substrate surface (Fig. 1a and b) and the growth of crystals which are parts of a polycrystalline structure (Fig. 1c and d) [14].

2.2.1. Growth of discrete crystals dispersed on the substrate

Crystals growing from the nuclei are randomly oriented due to the random orientation of the nuclei. The complete coalescence of the crystals touching each other is a grain coarsening resulting also in the development of discrete single crystals and connected to some changes in the orientation controlled mainly by the minimisation of the substrate–crystal interface energy. The intersection lines of the crystal side faces and the substrate present a structure precondition specific for the growth of these crystals. The intersection lines can be active or passive in the monolayer nucleation on the side crystal faces. In the first case, the movement of the monolayer growth steps proceeds from the intersection line to the top of the crystal, while in the second case, the movement of the growth steps proceeds to the direction of the intersection line. In the presence of impurities, the direction of the movement of the growth steps will be important in determining the location of the developing second phase, e.g., SCL.

2.2.2. Growth of crystals as parts of a polycrystalline structure

When a crystal becomes part of a polycrystalline structure, it might have various grain sizes and orientations as well as surface conditions. These structure conditions will

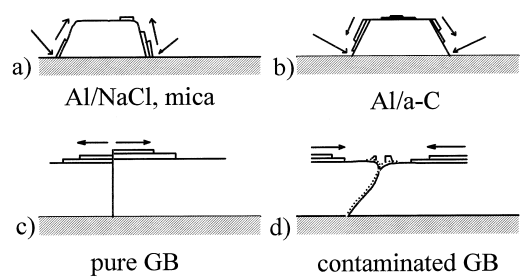

Fig. 1. Types of crystal growth in polycrystalline thin films: (a) and (b) growth of dispersed individual crystals on the substrate surface, the intersection line between the substrate and the side crystal faces are active (a) and passive (b) in the monolayer nucleation; (c) and (d) role of grain boundaries in the growth of crystals as parts of a polycrystalline matrix, pure grain boundary active in the monolayer nucleation (a), contaminated grain boundary passive in the monolayer nucleation (b).

determine its behaviour in the collective crystal growth characterising the polycrystalline film structure.

The growing faces of a crystal are parts of the free surface of the film. These crystal faces correspond to the equilibrium crystal shape and are determined by the orientation of the crystal at high purity conditions. A growth competition can start among the neighbouring crystals in case of different orientation according to the types of their growing faces, i.e., to their orientation [27–29]. The faster growing crystals will grow over the slower growing ones developing V-shaped crystal forms. This competition is terminated when only crystals exhibiting the same type of crystal faces proceed to the free surface. This competitive crystal growth represents an orientation selection among the crystals resulting in the so called *competitive growth texture* in the part of the film away from the substrate. The consequence of this competitive growth is the development of a changing morphology and texture along the film thickness. This is very pronounced at substrate temperatures where grain boundary migration is negligible. In that case, a small grained structure (corresponding mainly to the nucleation density) of random orientation exists in the substrate-near part of the film. It is followed by a part of V-shaped grains accompanied by an increase of the volume of preferentially oriented crystals. This process concludes later in the development of a columnar structure with a nearly unique crystal orientation.

The intersection lines of grain boundaries with the free surface can be active (this is the high purity case) or passive (this is the case of the contaminated grain boundaries) in the monolayer nucleation. Consequently, in the presence of impurities, the property of the grain boundaries will determine the surface area where the segregated SCL will develop [17].

2.3. Grain growth

In the course of structure evolution of polycrystalline films, two kinds of grain growth take place in the temperature interval where the grain boundary mobility has a considerable value. Grain growth takes place at first in the coalescence stage by complete coalescence of the contacting dispersed single crystal islands resulting in a new single crystal. Normal and abnormal grain growth can take place by grain boundary migration in polycrystalline islands and in continuous films.

The grain growth in case of complete coalescence results in a single crystal island on the substrate. The type of complete coalescence might be either a very fast, liquid like process, or a slower one. The type is determined mainly by the sizes of contacting islands, and by the conditions of the contacting surfaces at a given temperature. The complete coalescence of the contacting crystals is a periodic process during the film growth and, beside the increase of grain size, it results also in the changes of crystal orientations due to lowering the free energy of the developing crystals. This is a restructuration process developing stepwise a texture related to the lowest interface energy. This texture is called as *restructuration growth texture*.

It should be noted that the 'surface recrystallisation' has been introduced by Movchan and Demchishin [1] for the grain growth taking place by complete coalescence on the substrate surface. Unfortunately, this expression has not been used in its original sense consistently in the literature.

Upon liquid-like coalescence, a shrinkage of the coalescing crystals takes place developing bare substrate areas, i.e., places where the condensation proceeds by secondary nucleation. In that case, structures and phenomena of nucleation, island growth and coalescence stages can coexist. This shows up, e.g., in the coexistence of texture and random orientation in the diffraction pattern.

When the film becomes continuous, most of the crystals forming the polycrystalline structure have already participated in a series of complete coalescence during the preceding growth stages. This means that their orientation is already in or near to a preferred orientation corresponding to the minimum of interface and surface energy. Abnormal grain growth [30] takes place in the structure by grain boundary migration [6,7], the direction of which is controlled by the minimisation of the substrate–film interface and free surface energy. This means that the grain coarsening is connected to the strengthening of the preferential orientation. During abnormal grain growth, the grain size distribution is bimodal. When the abnormal grain growth can be completed during the film deposition, the as grown film has a monomodal grain size distribution. This might be realised at high T_s/T_m.

2.4. Formation of new phases by process-induced segregation of impurity species

Impurity species impinging onto the film surface can be adsorbed and segregated on the growing crystal faces or dissolved in the crystal lattice [8,31,32]. According to the experimental results, active impurities can inhibit the operation of the fundamental structure forming phenomena (decrease in grain size, weakening texture) (*inhibitor*) or promote [13,33] them (increasing grain size) (*promoter*). Those inhibitors have the strongest influence on the operation of the structure forming phenomena, i.e., on the structure evolution which are segregated by the atomic processes of the crystal growth and develop bidimensional covering layers on the growing crystal surfaces (SCL) and grain boundaries. Namely, SCL-s covering partly or completely the surfaces of the crystals and grain boundaries can control directly both the crystal growth and the grain growth, which are the most active phenomena of the structure evolution. Impurity segregation can also be related to the migration of grain boundaries [18,19].

The process-induced surface segregation of impurity species stems from the atomic processes of the monolayer

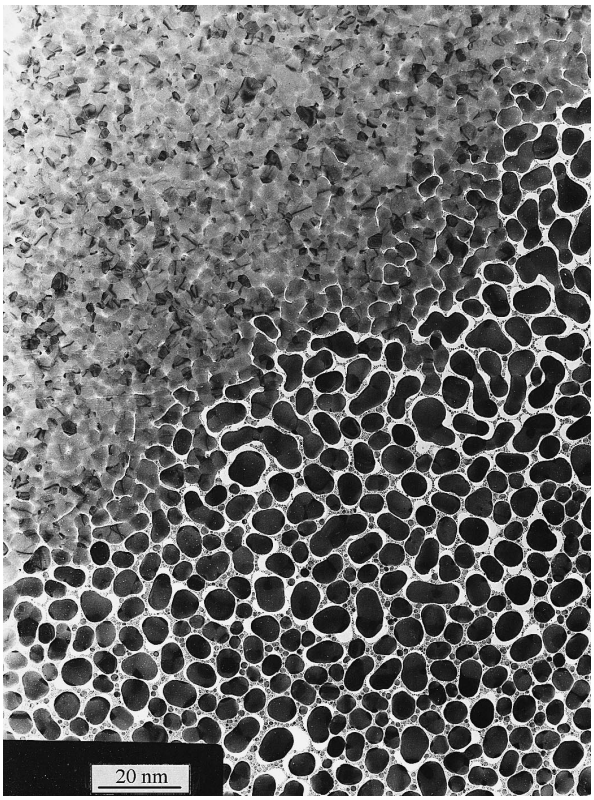

Fig. 2. Co-depositing Sn promotes the coalescence of Al crystals shown in the area marked by (B). Area marked by (A) is the pure Al film. 100-nm thick Al film deposited at room temperature, at 1 nm/s deposition rate on a-C layer supported by a TEM microgrid. Residual pressure was 10^{-4} Pa.

crystal growth discussed in details both in case of individual crystals dispersed on the substrate surface [14] and in case of polycrystalline structures [17] including the role of the activity of the intersection lines. The developing SCL limits the crystal growth on the covered surface areas resulting in the rounding and truncation of crystal shapes as well as in the appearance of dome-shaped surface features in continuous films. In case of complete surface coverage, the growth of crystals is fully blocked and the condensation proceeds by repeated nucleation. In that case, the thick films are composed of three dimensional (equiaxial, globular) overgrown crystals separated by the layers of the impurity phase. Consequently, thin films deposited under the influence of active impurities are two phase systems.

Impurity phase covering either the surface of individual islands or the grain boundaries in the polycrystalline structure can inhibit the grain boundary migration both at the coalescence and in case of grain coarsening. By this way, the processes developing the grain sizes and the restructuration texture are inhibited by the developing impurity phases and result in decreased grain sizes and weak texture or even in random orientation.

The surface chemical interaction of impurity species and, by this way, the segregation conditions might depend on the type of crystal face as in the case of aluminium and oxygen system [31,32]. In that case, the crystal growth is not inhibited on the crystal faces on which the impurity species are not segregated. Crystals with these types of growing faces can grow continuously in the polycrystalline structure, while the growth of others is inhibited. This is a good basis for the development of growth competition connected to orientation selection among the neighbouring crystals.

Experiments on vapour deposited Al/Sn and Al/In systems revealed that impurities can promote the operation of the fundamental structure forming phenomena by increasing the grain size at the same deposition parameters (Fig. 2) [13,33]. It turned out also that these additives decrease the inhibition effect of oxygen on the structure evolution of Al thin films [33]. Similar effect of additives have been found at electrolytic deposition of Au/Pb system [34].

3. Correlation between growth morphology and orientation of crystals

The analysis of the fundamental structure forming phenomena revealed that the evolution of the crystals morphology is related to the evolution of the crystals orientation through common mechanisms [35]. On this basis, textures can be correlated to the various zones of the structure zone model. On amorphous substrates, the nuclei are randomly oriented and the textures are evolutionary textures due to selection mechanisms. This selection can be related either to the competitive growth of crystals (due to the different growth rates in various crystallographic directions or to the crystal face dependent segregation of inhibitors) or to the abnormal grain growth controlled by the minimisation of the surface and interface energy. In the first case, the texture is *competitive growth texture*, while in the second case, it is the *restructuration growth texture*.

4. Main characteristics of structure zones and construction of new structure zone models

By analysing the structure zone models published in the literature [1–6], the various structure zones can be characterised by the following way (Fig. 3).

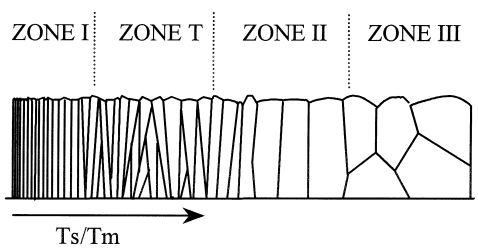

Fig. 3. Main characteristics of structure zones used in the literature [1–6].

In *zone I*, the film is composed of fibres of small diameter (1–10 nm) determined by the nucleation density and statistical fluctuation. The crystalline fibres grow out of the primary nuclei and proceed to the top of the film. The fibres are often collected into bundles. This is a rather homogeneous structure along the thickness of the film with increasing diameter of fibres by increasing T_s/T_m. The crystals contain probably high density of defects and the grain boundaries are porous. This structure belongs to the temperature interval $0 < T_s/T_m < 0.2$ where neither the bulk diffusion nor the self surface diffusion has remarkable value.

In *zone T*, the structure is inhomogeneous along the film thickness. It is fine crystalline at the substrate, composed of V-shaped grains in the next thickness range while can be columnar in the upper part of thick films. This zone belongs generally to the temperature interval $0.2 < T_s/T_m < 0.4$ in which the self surface diffusion is remarkable, however, the grain boundary migration is strongly limited.

Zone II represents a homogeneous structure along the film thickness composed of columns penetrating from the bottom to the top of the film. The grain boundaries are nearly perpendicular to the film plane. This zone is characteristic for high substrate temperatures ($T_s/T_m > 0.4$).

In *zone III*, the structure is characterised by equiaxed (globular) three dimensional grains, which is a direct indication that the crystal growth has been blocked periodically. This kind of structure is generally attributed to the high substrate temperature range. It will be shown, however, in the next paragraphs that the development of this kind of structure is the consequence of the presence of inhibitors and can exist at every substrate temperature with different grain sizes.

4.1. Concept of the basic structure zone model

The structure evolution in polycrystalline thin films can be discussed universally on the basis of the fundamental structure forming phenomena by considering the dependence of these phenomena on the temperature and on the actual structure preconditions existing in the film. Considering the basic structure forming phenomena (nucleation, crystal growth and grain growth), a basic structure zone model can be constructed. This model interprets the structure evolution at various temperatures and when no effects of co-depositing foreign species have to be considered.

Fig. 4 illustrates the structures at the various temperature intervals at increasing film thickness. In the temperature interval $0 < T_s/T_m < 0.2$, the nucleation density determines the lateral sizes of the fibres which are growing uninterruptedly side by side binding the adatoms at their impinging sites described by the ballistic models [36]. The orientation of fibres follows the random orientation of the nuclei. This is the structure of *zone I*.

In the temperature interval $0.2 < T_s/T_m < 0.4$, the surface diffusion has a considerable value. A structure attributed to *zone T* develops by competitive growth of differently oriented neighbouring crystals [28,29]. In the lower part of this temperature interval, no grain boundary migration can be expected. Therefore, the lateral size of competing crystals at the substrate is determined by the nucleation density. These crystals are in random orientation. At higher temperatures more and more grain boundaries become mobile resulting in the lateral growth of grains situated on the substrate. Due to this process, a weak preferred orientation develops strengthening gradually with temperature (Fig. 4b). The development of V-shaped crystals is a result of the competition taking place among the differently oriented neighbouring crystals (Fig. 4c) which concludes in a columnar morphology at higher thickness (Fig. 4d).

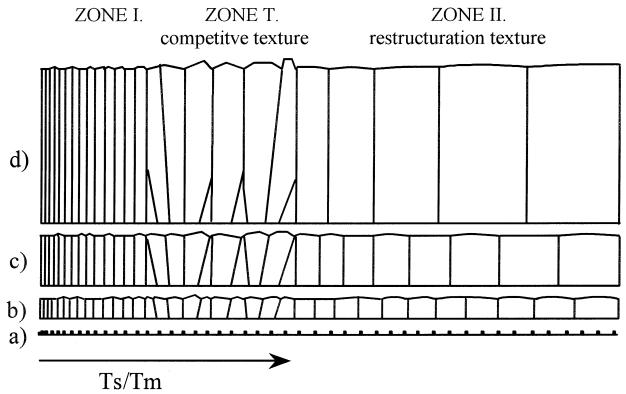

Fig. 4. Basic structure zone models at various film thickness.

The texture is changing with the thickness in this zone. In the lower temperature range, the substrate-near part shows random orientation, while moving away from the substrate, the competitive growth texture is increasing. In the higher temperature range, the substrate-near part shows a weak restructuration texture controlled by the surface energy minimisation. Moving away from the substrate, the restructuration texture can diminish and the competitive growth texture increases when these two are different.

In the temperature interval $0.4 < T_s/T_m$, the effect of grain boundary migration becomes decisive. The first structure of randomly oriented small grains is dissolved gradually by the coalescence and grain coarsening (Fig. 4a–d). This strong restructuration is controlled by the minimisation of the interface and surface energy and develops the *restructuration growth texture*. Being the grain boundaries mobile, the minimisation of the grain boundary energy can also take place resulting in grain boundaries perpendicular to the film plane. The film is composed of columnar crystals with similar orientation. The lateral size of the grains increases with increasing temperature.

4.2. Concept of the real structure zone models

In the case of real films, the co-deposition of active impurity species is hardly avoidable. The additional struc-

ture forming phenomenon, the process-induced segregation of active impurity species and the formation of SCL becomes active already at very low, sometimes hardly detectable impurity concentration of the impinging vapour beam.

The real structure zone models representing the temperature dependent structures at various levels of impurity concentration are given in Ref. [15]. In Fig. 5, the changes of the structure and the orientation conditions with increasing impurity concentration is shown for aluminium oxygen system deposited at room temperature, which corresponds to zone II in the basic structure zone model.

At low level of oxygen concentration, the grain growth in the coalescence stage is not limited remarkably, and the restructuration texture can develop. The growth of oriented grains develop the columnar structure according to zone II (Fig. 5a), however, the decreased mobility of contaminated grain boundaries limits the further grain growth. The width of columns (Fig. 5b) will be smaller than those according to the basic model, but the texture is still the restructuration texture, but weaker than in the case of the basic model.

At medium level of oxygen concentration, the grain growth is limited already in the coalescence stage. Because of the random orientation of the first crystals, a growth competition will start related to the different segregation property of oxygen on the different Al crystal faces [9,10,31,32]. A structure corresponding to zone T and competitive growth texture is developing (Fig. 5c).

At high level of oxygen concentration, the growth of crystals is blocked periodically by the developing SCL covering the whole surface of the crystals. It means also that no grain growth can take place hindering also the development of the restructuration growth texture. The film is composed of three dimensional equiaxial (globular) grains with random orientation according to zone III (Fig. 5d).

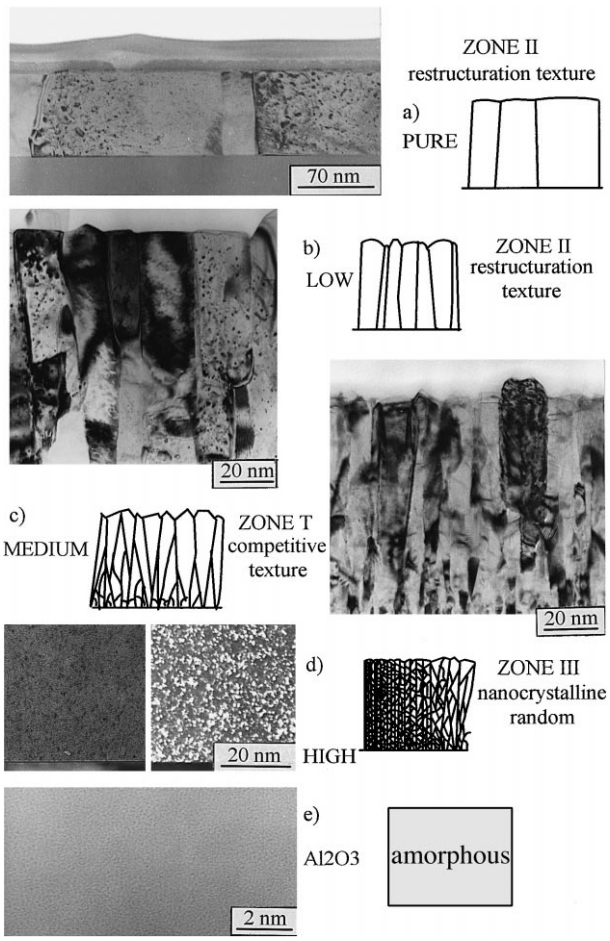

Fig. 5. Structural characteristics of Al thin films deposited by thermal evaporation at room temperature at normal high vacuum condition (a), at low (b), medium (c), high (d) and very high level of oxygen contamination.

5. Conclusions

The identification of the fundamental structure forming phenomena (nucleation, crystal growth and grain growth as basic phenomena and the process-induced segregation of impurity species as an additional phenomenon) present a good basis for a coherent view to understand and describe the structure evolution of polycrystalline films. This involves that when investigating the correlation between the applied technology (e.g., CVD, PVD including biased sputter deposition, ion assisted deposition) and parameters (e.g., ion energy, incidence angle) and the structural characteristics one has to analyse and understand how the technology and parameters alter the structural preconditions and the kinetics of the structure forming phenomena related to the basic structure zone model.

Considering only the basic structure forming phenomena, the basic structure zone model can be constructed which describes the structure evolution of pure metal films on amorphous substrates. Taking the additional phenomenon, the process-induced segregation of co-depositing impurity (additive) species into account, the structure zone model of real polycrystalline films can be constructed.

The correlation between the grain morphology and texture of the films can be understood on the basis of the fundamental structure forming phenomena. This correlation made possible to complete the structure zone models by relating the growth textures to the specific zones.

It has been recognised that impurities or additives can inhibit or promote the operation of the fundamental structure forming phenomena. This presents new possibilities for tailoring stabilised artificial (nanocrystalline or extremely large crystalline) thin film structures and to interpret the results published recently on the role of parameters in the development of nanocrystalline structures [37,38] and in changing the grain size and texture [39–41].

Acknowledgements

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